



ashwa MOBILITY
FOUNDATION

LIVE IT. LOVE IT **RACE IT.**

EUROPEAN HYPERLOOP WEEK EDITION

**APRIL 2025
EDITION**

**FROM
DESIGN TO
DEMO,
MAKING
HYPERLOOP
HISTORY**

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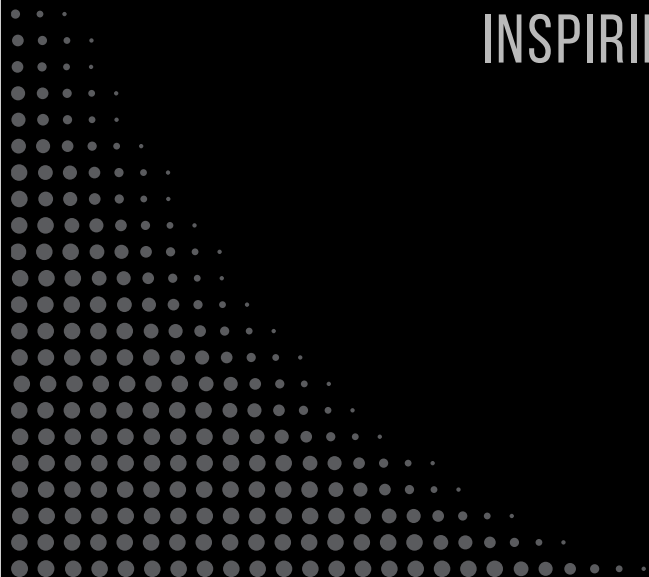


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**VISION
AND
MISSION**

GUIDING PURPOSE,
INSPIRING FUTURE.



> MISSION AND VISION <

VISION

- To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.
- To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

- Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.
- Showcase India's engineering expertise in mobility on a global stage and provide talent opportunities for the industry.
- Develop cost-effective, sustainable transportation solutions for the masses.



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ENGINEERING TOMORROW

ASHWA MOBILITY'S JOURNEY AT
EHW 2024





ASHWA MOBILITY AT EUROPEAN HYPERLOOP WEEK

Ashwa Mobility, the pioneering student-led innovation team from RV College of Engineering, marked a significant milestone in its Hyperloop journey with an impactful debut at the European Hyperloop Week (EHW) 2024. Hosted in Zurich, the event gathered leading minds and future-shaping technologies from across the globe, and Ashwa's presence was both prominent and promising.

Having launched their Hyperloop project in 2019, Ashwa has consistently demonstrated technical prowess, making it to the finals in past editions of EHW. However, 2024 was a turning point, as the team presented a fully functional prototype, stepping beyond theoretical design into the realm of practical implementation. This marked the culmination of years of research, testing, and unwavering commitment to sustainable high-speed transportation.



Team Ashwa representing India

"Being the Team Captain of the Hyperloop Project at Ashwa was the most exciting part of my college life. I had the privilege of leading the team in finalizing designs and manufacturing our first Hyperloop prototype from scratch all within a single year. It was a rollercoaster of dropped ideas, realigned goals, tight budgets, shifting timelines, and tough decisions. But it all paid off as we ended the season representing our college and country at the European Hyperloop Week in Zurich and the Parivahan Conference at IIT Madras and more importantly, with countless unforgettable nights spent discussing, debating, and engineering everything Hyperloop." - Ayush Nistala, Team Captain, Hyperloop.

The event kicked off with an introductory session followed by an open day, where Ashwa Mobility engaged with delegates and fellow Hyperloop teams. These interactions offered valuable insights into global design approaches and gave the team a fresh perspective on refining their own mechanical systems.



Ashwa members interacting with Delft Hyperloop and Hyperloop experts

“Being a part of the Ashwa Hyperloop team has been one of the most defining experiences of my college life. From brainstorming sessions in the early stages to tackling real-world engineering challenges, every moment taught me something new—not just technically, but about collaboration, perseverance, and leadership. I was lucky to work alongside an incredibly passionate and talented group—my batchmates who constantly pushed boundaries, and our juniors whose enthusiasm brought fresh energy to the team. One of the most memorable highlights was representing Ashwa at the European Hyperloop Week. Seeing our pod alongside teams from across the globe was surreal, and it reinforced how far we had come as a student-led team. The journey was tough, but the growth it brought was unmatched. I’ll always carry this chapter with a deep sense of pride and gratitude.” - Sumeet, Chief Engineer

On Day 2, Ashwa Mobility attended their first expert talk, gaining insights into emerging Hyperloop innovations. They showcased their mechanical subsystem in the finals and participated in a workshop, engaging with international teams to enhance their technical and collaborative skills. Day 3 featured another expert talk and the finals for the Traction Subsystem, where they demonstrated their innovative drive technology. The team also attended a second workshop and gained valuable exposure through presentations by leading teams like Hyperloop UPV and Avishkar Hyperloop.

On Day 4, Ashwa Mobility participated in their third Hyperloop workshop, where their innovative ideas earned them a win one of the week's key highlights. They also attended a presentation by Swiss Loop, gaining further inspiration and insights from one of Europe's top teams. Day 5 was dedicated to the first round of technical inspections, during which Ashwa Mobility's pod underwent detailed scrutineering of its mechanical and sense & control systems, along with a successful mechanical push test to validate operational readiness. On Day 6, live demonstrations by top teams such as Hyperloop UPV, Mu Zero, Swiss Loop, and Delft Hyperloop provided valuable comparative insights, while judges offered constructive feedback on improving the mechanical design of their pod. On the final day, Ashwa presented their work to the general public in Zurich, engaged with dignitaries, and proudly represented RVCE and India on an international stage.



Team Ashwa showcasing engineering innovation and collaboration

"EHW was my first event where I got the opportunity to represent Ashwa Mobility and my country at a international conference.

I had no idea what we should do going there as I was in 4th semester.

When we reached there, we entered the event venue. I saw a team with huge tents , prototype and each team consisting of 50 members and we were just a team of 8.

Then we spoke to other teams, they were very nice and they told about their project. We would go around the city after the event and spend time with other teams.

I made a good friend there, he is from Spain and we talked about each others team and culture. I gained good contacts and got to know how different teams work- the challenges they faced and how they solved it." - Hanish Verma



Key Highlights of the event

Ashwa Mobility was nominated in the finals for Best Subsystem Design in both the Traction and Sense & Control categories, acknowledging the team's technical excellence.



Presentation by Ashwa at the Mechanical and S&C subsystem finals

Ashwa engaged with leading global experts working on real-world Hyperloop applications, gaining valuable perspectives on scaling the technology for future generations.

Through interactions with teams from around the world, Ashwa gathered crucial insights into cutting-edge design practices and future development plans, accelerating their own innovation process.

“Participating in the European Hyperloop Week was a defining moment during my time at Ashwa. The event offered deep exposure to cutting-edge technologies and operational strategies. Working late nights as part of the sponsorships team was challenging yet rewarding which was as memorable as it was hectic and it was an honor to contribute to a project built on years of effort by talented Ashwa members.” - Ravikiran, Sponsorship, Marketing and PR.

The team won a prestigious Hyperloop Workshop, where they proposed innovative strategies for integrating Hyperloop technology into large metropolitan areas, while also emphasizing its potential to foster international collaboration and diplomacy.



Ashwa's Participation in Hyperloop Workshops and subsequent victory

By participating in EHW 2024, Ashwa proudly represented R.V. College of Engineering and India on a global platform, reinforcing their commitment to advancing sustainable transportation technology.

“EHW 2024 at ETH Zurich was one of the most transformative and memorable experiences of my college life. As a junior in my 4th semester, I initially felt unsure of my abilities and wondered if I could truly contribute to something this big. The intense late nights at the workshop in the weeks leading up to the event taught me perseverance, attention to detail, and how to stay composed under pressure. Stepping into an international environment, I was struck by how much there was still to learn—an eye-opening realization that sparked both humility and motivation. One of the most rewarding aspects was learning about the financial side of engineering projects from Arunraj Subbaraj (Founder, Ashwa Mobility) and Dr. Ravindra S Kulkarni sir. Their clarity, structured thinking, and approach to managing resources taught me lessons that went far beyond the classroom. I’m thankful to my senior teammates for trusting me and helping me grow through this journey.” - Venkat Shreyas

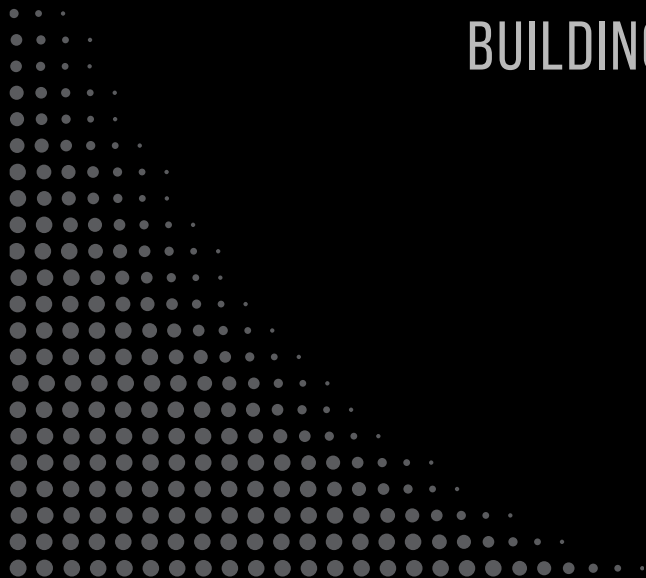


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ALUMNI CORNER

CONNECTING MEMORIES,
BUILDING FUTURES!



ALUMNI CORNER



Ipparla Ashakiran, Project Manager, Hyperloop

Ashakiran played a key role in Ashwa Mobility's Hyperloop project as the Head of Brakes and Logistics. He was responsible for developing the pod's braking system and managing logistics, ensuring smooth transport and setup during European Hyperloop Week 2024. His efforts were crucial to the team's overall success at the event.

My time at Ashwa Mobility Foundation was foundational in building both my technical and soft skills. Being the sole member responsible for the brakes subsystem, I had to handle everything from design and calculations to manufacturing and testing. This experience taught me how to work independently under pressure, prioritize effectively, and maintain a strong attention to detail—skills that proved invaluable at EHW 2024. Hyperloop systems are inherently interdisciplinary and fast-paced, and Ashwa gave me the confidence to contribute meaningfully in such an environment, especially when collaborating with diverse teams and working toward tight deadlines and also very happy to have good team mates who supported all decisions in tough times.

We decided to move away from our previous brake system design and develop a new version that would better suit our Hyperloop prototype. After discussing several ideas within the team, everyone unanimously supported my proposal for a frictionless braking system. That's where the real challenge began.

Coming from a mechanical engineering background, I found it initially very difficult to start the design process, especially since I had no prior experience with validating electromagnetic concepts or using simulation software. However, I never lost my motivation. I relied on my foundational knowledge of physics from school and College and began diving deep into electrical engineering textbooks and journals on electromagnetism. I also reached out to professors for guidance, which helped me build an initial concept.

To validate the design, we explored various software options and finally settled on EMWorks—huge thanks to their team for their incredible support. Once we had a validated design, we moved on to the manufacturing stage. Sourcing the right components proved to be another major hurdle, but we reached out to other teams for vendor suggestions, and with persistence, we finally managed to build the braking system.

Unfortunately, we couldn't test the system in time, which was one of the reasons we didn't score in that segment at European Hyperloop Week 2024. On top of that, managing the logistics was extremely demanding. I'm especially grateful to ArunRaj, our team coach, who played a crucial role in ensuring our participation. He supported me through vendor communications, documentation, and constant guidance.

In the end, it was a proud and unforgettable experience to participate in an international competition, representing our country, college, and team. The journey—from the initial idea to presenting it on a global stage—was filled with learning, motivation, and growth.

Participating in the event gave me a significant boost in confidence. The feedback from the judges, along with observing the advanced technologies used by other teams, was incredibly insightful and educational. The exposure to such a competitive environment helped me stay calm and focused during Formula Bharat 2025, allowing me to contribute effectively to achieving the goals of our subsystem without feeling overwhelmed. Although we couldn't end the event on a high note—which was a bit disappointing—the entire experience played a major role in shaping my career path and helped me grow into a more capable and well-rounded individual.



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**DRIVEN BY
SUPPORT**

A TOKEN OF THANKS TO OUR
SPONSORS





A TRIBUTE TO OUR SPONSORS

SPONSOR APPRECIATION DRIVE 2025

As a gesture of heartfelt gratitude, Ashwa Mobility embarked on a Sponsor Appreciation Drive to personally distribute certificates of appreciation to all its valued sponsors. These organizations have been powering Ashwa's journey through innovation, competition, and continuous development. By acknowledging their generous support, the team aimed to strengthen ties and celebrate the collaborative spirit that fuels every milestone at Ashwa.



GOWRISHANKAR ENGINEERING INDUSTRIES

Gowrishankar Engineering Industries was established in 1985 as a proprietorship Business. Started with a vision to utilise the expertise gained over the years in the automobile industry, the IATF 16949 certified company has for almost three decades served the ever growing automobile sector with quality products.



AKSHAYA MOTORS



Akshaya Motors is a wholly-owned company of the Advait Group. We are one of the largest automotive retail chains in India that's focused on delivering a world-class customer experience across all business functions. Akshaya Motors -The Fastest Growing Mercedes-Benz Franchise Partner with Showrooms in Bangalore (60,000 sq. ft.) Mysore & Hubballi.



AART MOTORSPORT HARMONIXX

They specialize in ECU and TCU remaps, intake and exhaust upgrades, turbo upgrades, and in-house dyno testing to optimize vehicle performance. Based in Bangalore, their expert team offers tuning services for cars across India, ensuring enhanced power and efficiency tailored to vehicle's needs.



FASTOLEX



Distributors and stockist of UNBRAKO fasteners Distributors and stockist of Raajratna stainless steel 304/316 fasteners, MFG of dowel pin and SS cable tie. MFG of turning components, Distributors and stockist of Hittco tools



MERITOR

MERITOR

Meritor is a global leader in drivetrain, braking, and mobility solutions for commercial vehicles and industrial markets. It provides axles, brakes, and suspension systems, serving industries like trucks, trailers, and defense.





SINEWAVE

**SINEWAVE GENERATORS
PRIVATE LIMITED**



Sinewave Generators Pvt Ltd is a leading provider of diesel generator solutions in Bangalore. As an authorized Cummins dealer, they offer sales, installation, and maintenance services along with rental options for high-capacity industrial generators. Their vehicle mounted generators come equipped with acoustic enclosures, are well-maintained and regularly serviced to ensure maximum reliability.



INFINEON

Infineon Technologies is a German semiconductor manufacturer specializing in microcontrollers, power electronics, and integrated circuits. They provide solutions for industries such as automotive, industrial, and consumer electronics, focusing on energy efficiency and security.



TRIUMPH LASER



Triumph Lasers is a laser cutting service provider based in Peenya 2nd Stage, Bengaluru. They offer precision laser cutting for various materials to meet industrial needs.



ARUANI GRID

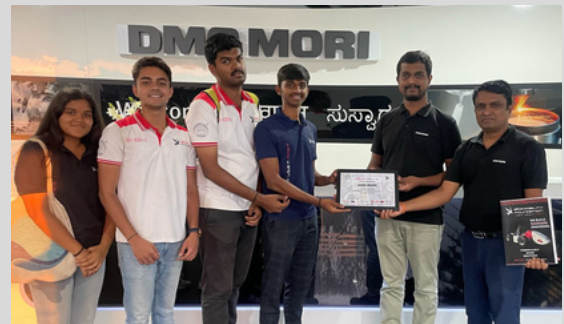


Aruani Grid provides a world-class racing experience for enthusiasts of all age groups, featuring an array of tracks that cater to novice and professional racers alike. It is not just a track; it's a breeding ground for champions, offering specialised training programs in go-karting and other motorsports disciplines.

DMG MORI

DMG MORI

DMG MORI is a leading global manufacturer of high-precision machine tools, specialising in technologies such as turning, milling, grinding, boring, ultrasonic, laser, and additive manufacturing. The company offers innovative services throughout the entire life cycle of machine tools, including automation and digitisation solutions.



JOE'S GARAGE



A 2 Wheeler workshop situated in Thimmiah Rd, Shivajinagar, Bengaluru which deals in repairing, tuning & even customising premium bikes & super bikes.

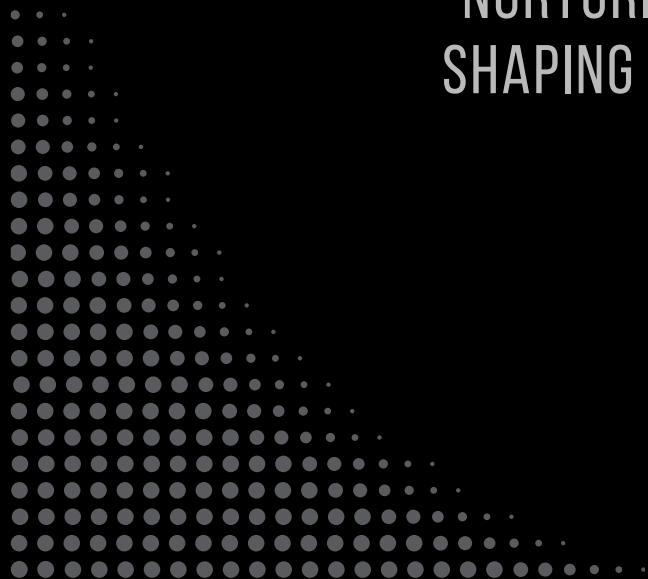


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TECHNICAL INSIGHT

NURTURING TALENT,
SHAPING THE FUTURE.





DRIVING DATA: BUILDING A CAN-BASED DISPLAY SYSTEM AT ASHWA RACING

The work one can do at Ashwa is as varied as one wants. It can vary from validation and testing of electrical circuits, approaching major corporations regarding components and sponsorships, dealing with customs, training juniors, doing cost analysis etc. All this and more can be done at Ashwa Racing without any constraints to learning!

The learning never stops before or after a batch goes to competition, since improvements are constantly made in iterations. This as an example can be said for the CAN based TFT LCD display that was implemented for post season testing of 2025 to test the functionality and extent of customization that can be done for the same.

A 3.5" inch TFT LCD display with the ILI9486 controller was used, and simple and essential values of the gearing and the RPM of the engine were displayed. But how does the microcontroller controlling the display get the information that continuously changes from the Engine (the beloved GSXR 600)?

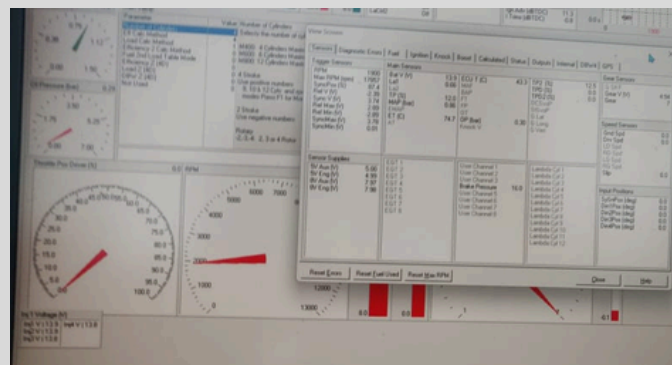
This is where the MoTeC M800, gold box ECU provided by MoTeC comes into play! The MoTeC M800 is called an ECU or Engine Control Unit, and is basically the brain of the entire car. Interfaced with all the sensors ranging from MAP to Brake Pressure sensors, it is able to observe the entire car and presents the entire information that it is provided in the form of hexadecimal format over STD CAN 2.0.



TFT LCD Display currently implemented for testing

CAN or, Controller Area Network is a differential protocol made by Bosch that is the industry standard for communication protocols over automotive solutions due to its reliability and adaptability for varying priority levels of sensors, stringent error checking capability and even a varying data size from 8 bytes to up to 64 bytes with CAN FD ! All this provides immense amounts of development options over the years, and hence why CAN is used everywhere.

However this huge amount of features are restricted to hardware only, that is CAN requires a CAN controller which deciphers the information which can be used by the user as well as a CAN transceiver that is used to quantify the differential voltages in the CAN bus to something that the CAN controller can use. Also, since CAN is a communication protocol and is fast enough where reflectance has a huge impact on the amount of correct bits received, we require termination resistors (120 ohms as specified by both MoTeC & TI) on the CAN bus to ensure impedance matching



Motec ECU manager being used to observe sensor data

Coming back to the point of the display, MoTeC is configured for using STD CAN 2.0 and the easy to use MoTeC ECU manager was used to setup the 11 bit identifier or address, along with the RPM and gear in the user defined list of sensor data to be sent over CAN, and flashing the configuration to the MoTeC ECU we start receiving information over the CAN bus with all the information with the address so that one can identify if it is the correct sensor that they need information from. This is then followed with DLC and then 2 bytes or 4 hex digits for each channel/sensor configured, which then can be byte shifted by 8 and added up to the correct decimal value. All this information is then sent to the display via the microcontroller, and which controls the entire logic of refreshing the display.

However, this would not have been possible without the help from seniors and super-seniors and alumni that even after countless doubts asked are more than happy and excited to clear doubts when one asks for it. This is the true philosophy of Ashwa Racing since its inception

*Signing out,
Aditeya Sarkar,
Electrical Member,
'27 Batch Ashwa Racing.*



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SPONSORS
(2024-2025)**

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DRIVING US TO GLORY!



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**TEAM
DIRECTORY
2025**

DRIVEN BY PASSION, UNITED BY ENGINEERING—
THIS IS ASHWA RACING!

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